

Answers:

1. True/False: The AND operator '&&' requires both operands to be true to return true.
 - a. True
2. True/False: The negation operator '!' flips a boolean value.
 - a. True
3. True/False: The OR operator '||' requires both operands to be true to return true.
 - a. False
4. True/False: What is the result of true && false?
 - a. False
5. True/False: What is the result of true || false?
 - a. True
6. True/False: What is the result of !true?
 - a. False
7. Write a code snippet to evaluate if a player can attack in a game. The conditions are: The player must not be reloading and must have ammo. Use boolean variables isReloading and hasAmmo.

```
bool canAttack = !isReloading && hasAmmo;
```

8. Write code to evaluate if a character should take damage. The character takes damage if their shield is down ('shieldDown') or if the attack bypasses shields ('bypassShields').

```
bool shouldTakeDamage = shieldDown || bypassShields;
```

9. You're designing a game where a player can only enter a dungeon if they are level 10 or higher ('playerLevel') and have a key ('hasKey'). How would you model this condition using boolean logic?

```
if (playerLevel >= 10 && hasKey) {  
    enterDungeon();  
}
```

10. In a first-person shooter game, write the logic to determine if a player can use a special ability. They can use it if their energy is full ('energyFull') or if they have an "Ability Boost" item ('hasAbilityBoost').

```
if (energyFull || hasAbilityBoost) {  
    useSpecialAbility();  
}
```

11. In an rpg game, a unit should be promoted if it has more than 100 experience points ('xpPoints') and is not already at maximum rank ('isMaxRank'). Write a code snippet to model this.

```
if (xpPoints > 100 && !isMaxRank) {  
    promoteUnit();  
}
```

12. In a survival game, a player character should eat if their energy level is below 20% or if 12 hours have passed since the last meal. Write a code snippet to model this logic.

```
if (energyLevel < 20 || hoursSinceLastMeal >= 12) {  
    eat();  
}
```